

GRADE 9 MATHEMATICS · MEASUREMENT & GEOMETRY

Surface Area & Volume

THE BIG IDEA

Surface area measures the
measures the space inside,
units.

SLANT HEIGHT VS. PERPENDICULAR HEIGHT

Use **perpendicular height (h)** for volume.

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skin, in cm^2 . **Volume**
in cm^3 . Never mix the

Use **slant height (l)** for surface area of pyramids

$$l^2 = r^2 + h^2$$

WORKED EXAMPLE

Cone: $r = 3 \text{ cm}$, $h = 4 \text{ cm}$

Slant height: $l = \sqrt{(3^2 + 4^2)} = \sqrt{25} = 5 \text{ cm}$

$SA = \pi r^2 + \pi r l = 9\pi + 15\pi = 24\pi \approx 75.4 \text{ cm}^2$

$V = \frac{1}{3}\pi r^2 h = 12\pi \approx 37.7 \text{ cm}^3$

⚠ WATCH OUT FOR

MISTAKE

Using slant height in the volume formula

MISTAKE

cm^2

FIX

Volume always uses **perpendicular** height

FIX

"Cone volume is half the cylinder"

MISTAKE

Forgetting to subtract hidden joined faces

 SELF-CHECK

Can you state all five SA and V formulas with
r and h?

It's **one third** the cylinder (same base and height)

FIX

Every joined face disappears from **both** shapes

out looking? Can you find slant height from